Problem Space Exploration

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Math Formula Derivations

Given a one-factor second order model, we derive a formula to compute SPV. Recall the second order model expressed in scalar form:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x^2 + \epsilon$$

which can be generalized to

$$\hat{\mathbf{y}} = \mathbf{F}(\mathbf{X})\beta + \epsilon$$

in matrix-vector form. Now we derive the SPV formula for an arbitrary 3-trial experimental setting. That is,

$$\mathbf{x} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

where a, b, and c represent factor levels for the univariate experiment. Now the model matrix is

$$\mathbf{F}(\mathbf{X}) = \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}.$$

To compute the SPV for this arbitrary model matrix, we need to compute

$$SPV(\mathbf{x}^T|\mathbf{X}) = N f^T(\mathbf{x}^T)(\mathbf{F}(\mathbf{X})^T\mathbf{F}(\mathbf{X}))^{-1} f(\mathbf{x})$$

so to make the calculation practicable lets make the inverse information matrix arbitrary:

$$(\mathbf{F}(\mathbf{X})^T\mathbf{F}(\mathbf{X}))^{-1} = \begin{bmatrix} A & B & C \\ D & E & F \\ G & H & I \end{bmatrix}.$$

Now for the one-factor setting we have

$$SPV(\mathbf{x}^T|\mathbf{X}) = 3 \times \begin{bmatrix} 1 & x & x^2 \end{bmatrix} \begin{bmatrix} A & B & C \\ D & E & F \\ G & H & I \end{bmatrix} \begin{bmatrix} 1 \\ x \\ x^2 \end{bmatrix}$$

$$= 3(A + x(D + B) + x^2(C + E + G) + x^3(H + F) + Ix^4)$$

The two factor setting is

$$SPV(\mathbf{x}^T|\mathbf{X}) = 3 \times \begin{bmatrix} 1 & x_1 & x_2 & x_1^2 & x_2^2 & x_1x_2 \end{bmatrix} \begin{bmatrix} f_{11} & \cdots & f_{16} \\ \vdots & \ddots & \vdots \\ f_{61} & \cdots & f_{66} \end{bmatrix} \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_1^2 \\ x_2^2 \\ x_1x_2 \end{bmatrix}$$

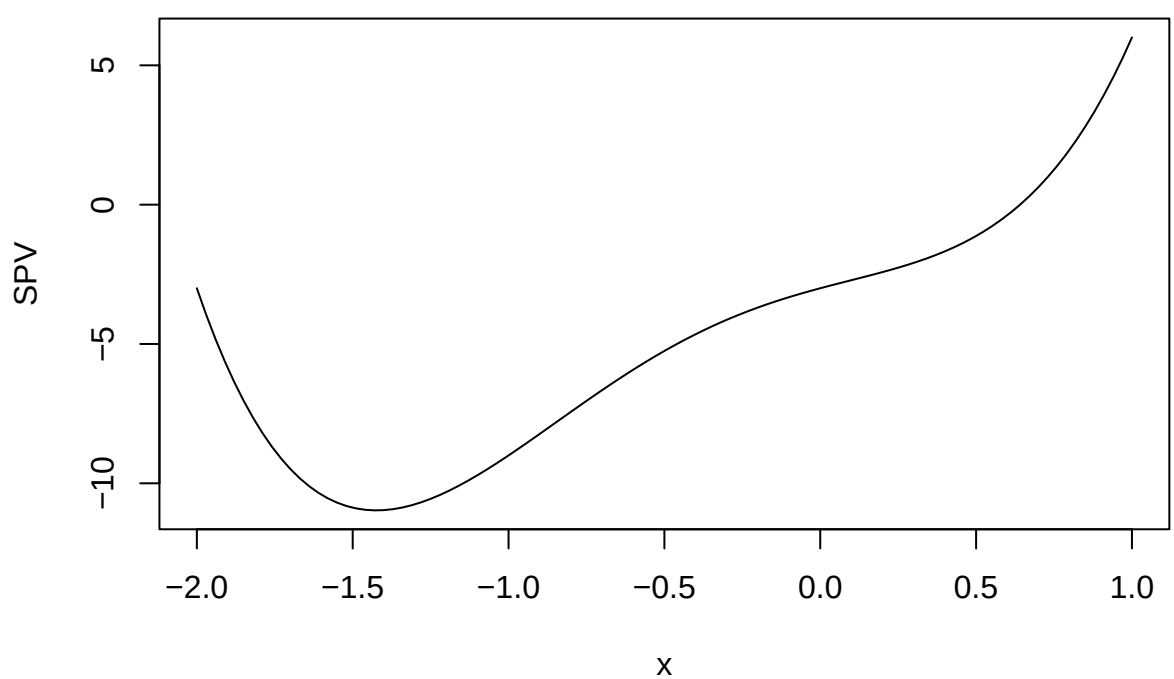
$$= \begin{bmatrix} \vec{x}^T\vec{f}_{1i} & \vec{x}^T\vec{f}_{2i} & \vec{x}^T\vec{f}_{3i} & \vec{x}^T\vec{f}_{4i} & \vec{x}^T\vec{f}_{5i} & \vec{x}^T\vec{f}_{6i} \end{bmatrix} \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_1^2 \\ x_2^2 \\ x_1x_2 \end{bmatrix}$$

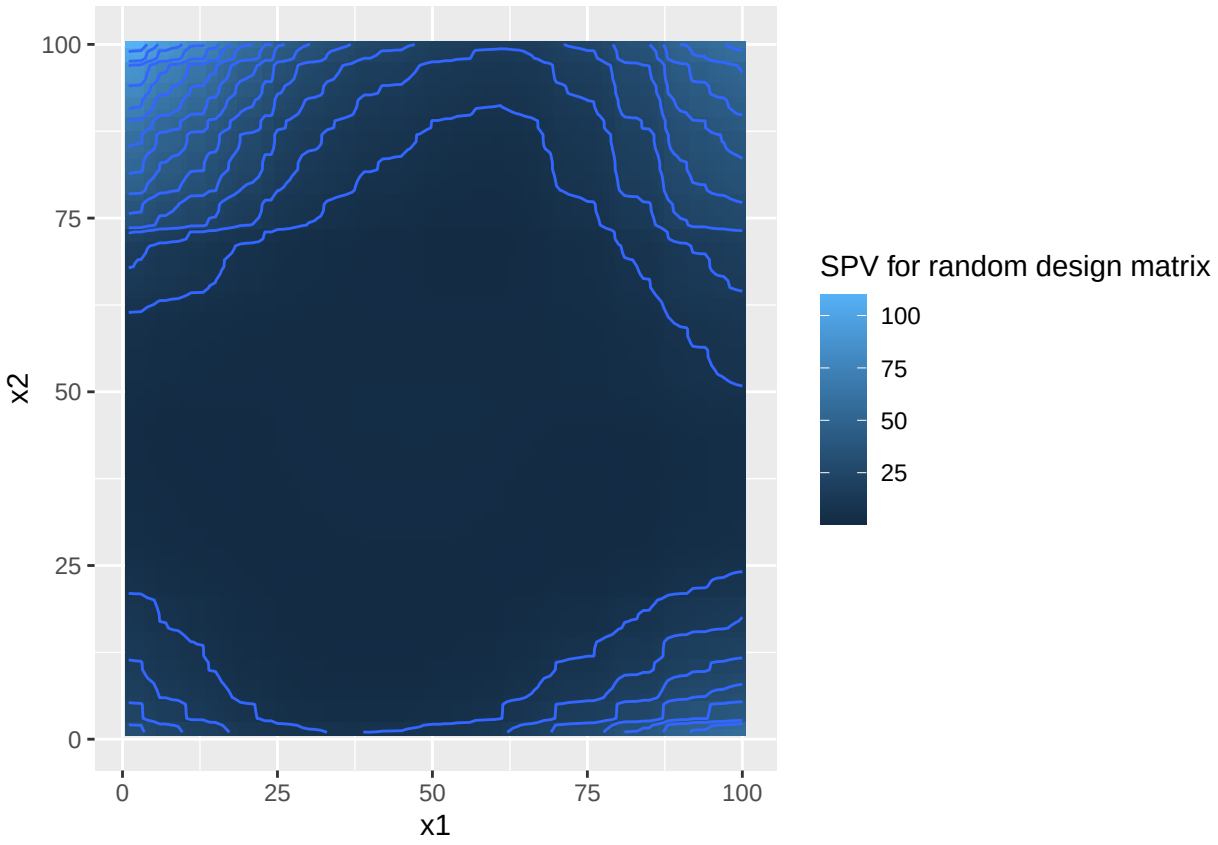
$$= \vec{x}^T\vec{f}_{1i} + \vec{x}^T\vec{f}_{2i}x_1 + \vec{x}^T\vec{f}_{3i}x_2 + \vec{x}^T\vec{f}_{4i}x_1^2 + \vec{x}^T\vec{f}_{5i}x_2^2 + \vec{x}^T\vec{f}_{6i}x_1x_2$$

The highest order terms in this expression will be x_1^4 and x_2^4.

Plotting SPV

For the K=1 case (and an arbitrary inverse information matrix):



Here's a similar plot for the K=2 case. For this problem the candidate design matrix is arbitrary.

```
##
## Attaching package: 'reshape'

## The following objects are masked from 'package:reshape2':
##
##     colsplit, melt, recast

## Warning: The dot-dot notation (`..level..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(level)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```